CprE 458/558: Real-Time Systems

Real-Time Networks – WAN
Channel establishment

QoS metrics: types of constraint

Metrics and their nature

- End-to-end delay: additive metric, path metric
- Delay jitter: additive metric, path metric
- Bandwidth: convex (min), link metric
- Packet loss: multiplicative, path metric

Theorem: A problem with two or more path constraints is NP-Complete, assuming the constraints are independent and the values are real.

QoS routing problem instances

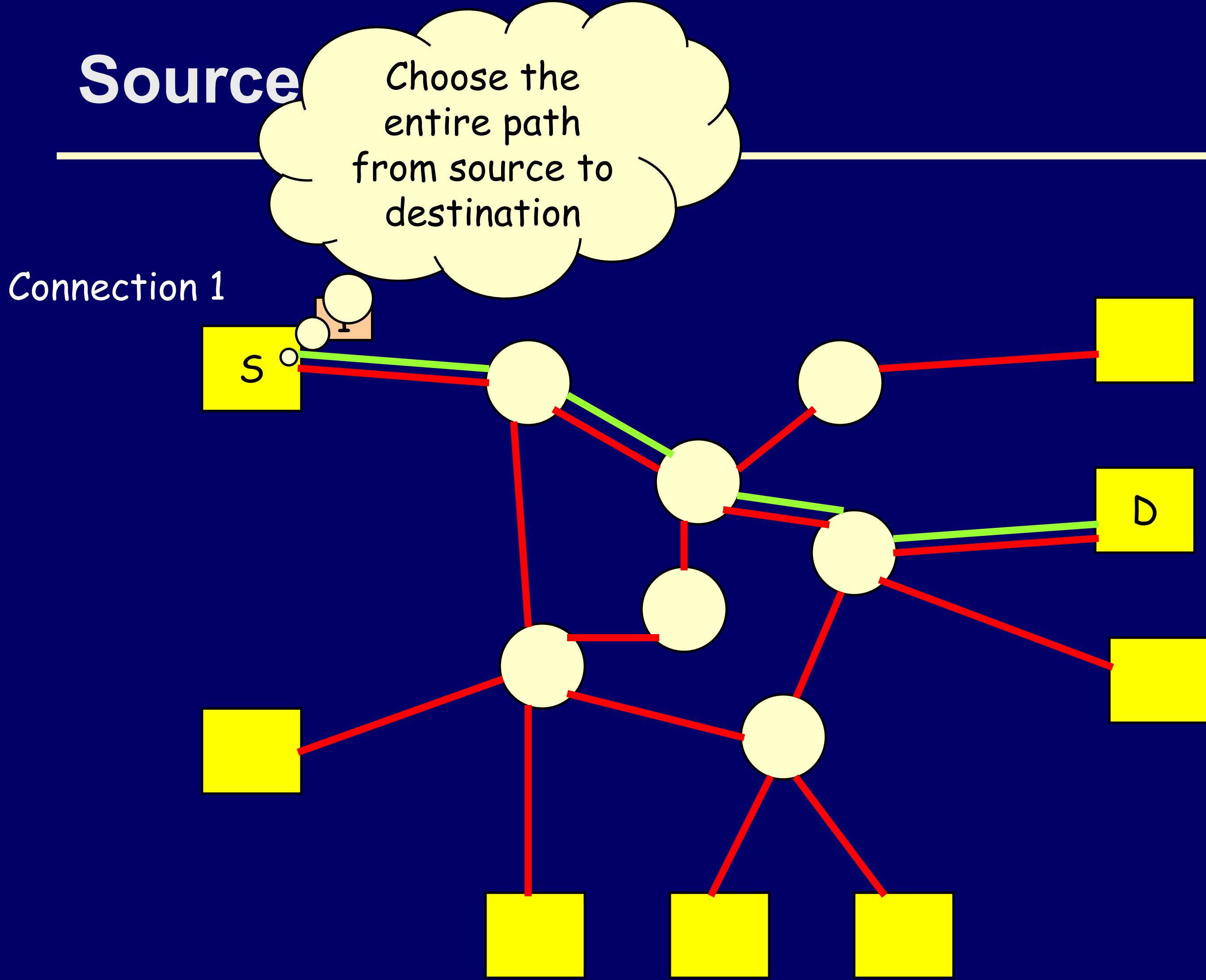
Given a graph G(V,E), with each edge labeled with (cost, delay, available bw).

- Bandwidth-constrained least-cost routing
  - Polynomial time problem
  - Solution: Remove edges that don't satisfy BW constraint and then run shortest path algorithm on the reduced graph.
- Delay-constrained, BW-constrained, least-cost routing
  - NP-complete problem
  - Heuristics exists

Channel establishment – QoS routing

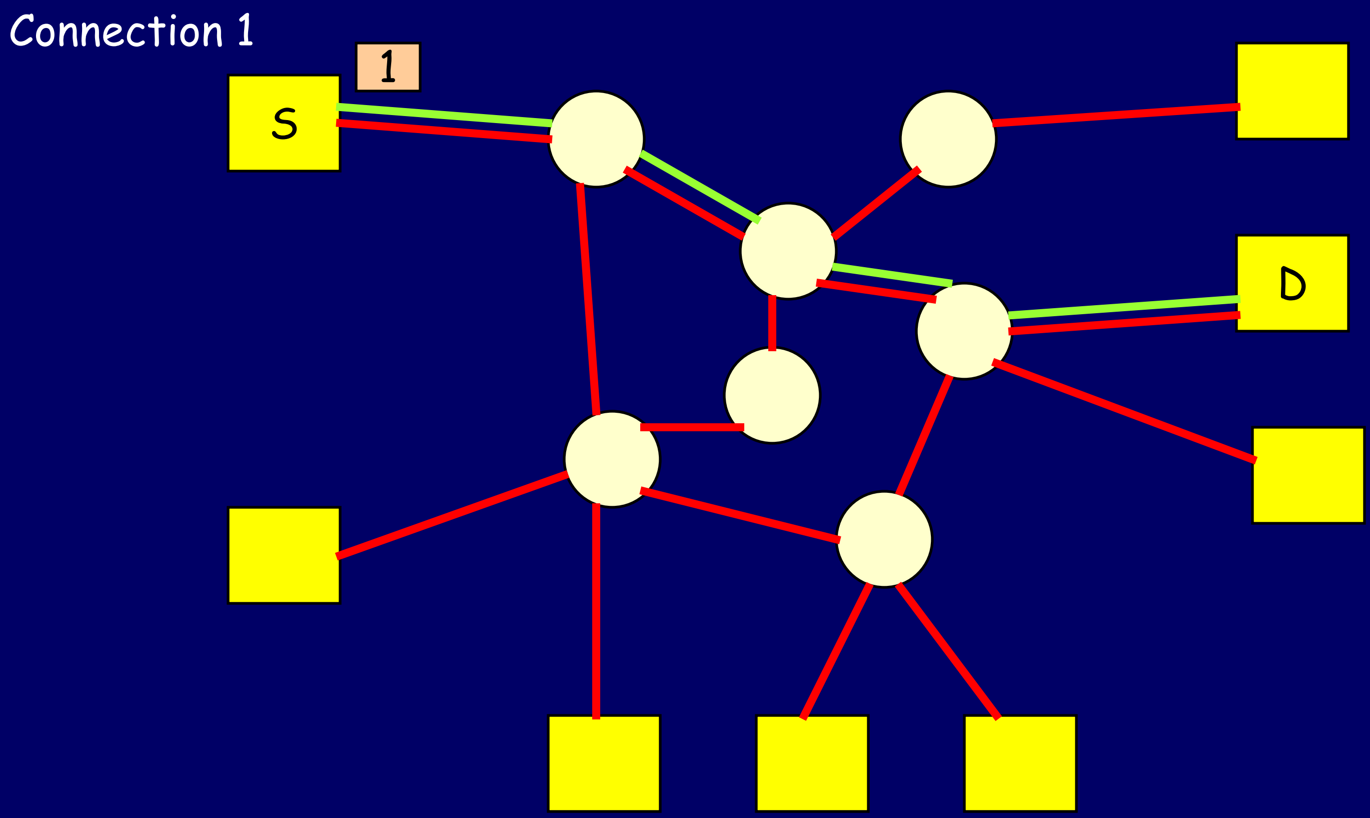
- Source Routing
  - Source computes the routing path
  - Obtaining global state is difficult
  - Not scalable
- Distributed Routing (hop-by-hop)
  - Each node decides what is the next hop
  - Lack of global knowledge → inferior paths
  - Scalable
- Hierarchical Routing
  - Based on aggregate state information
  - Scalable (works well for inter-domain routing)
  - Compromise between source and distributed routings

Source

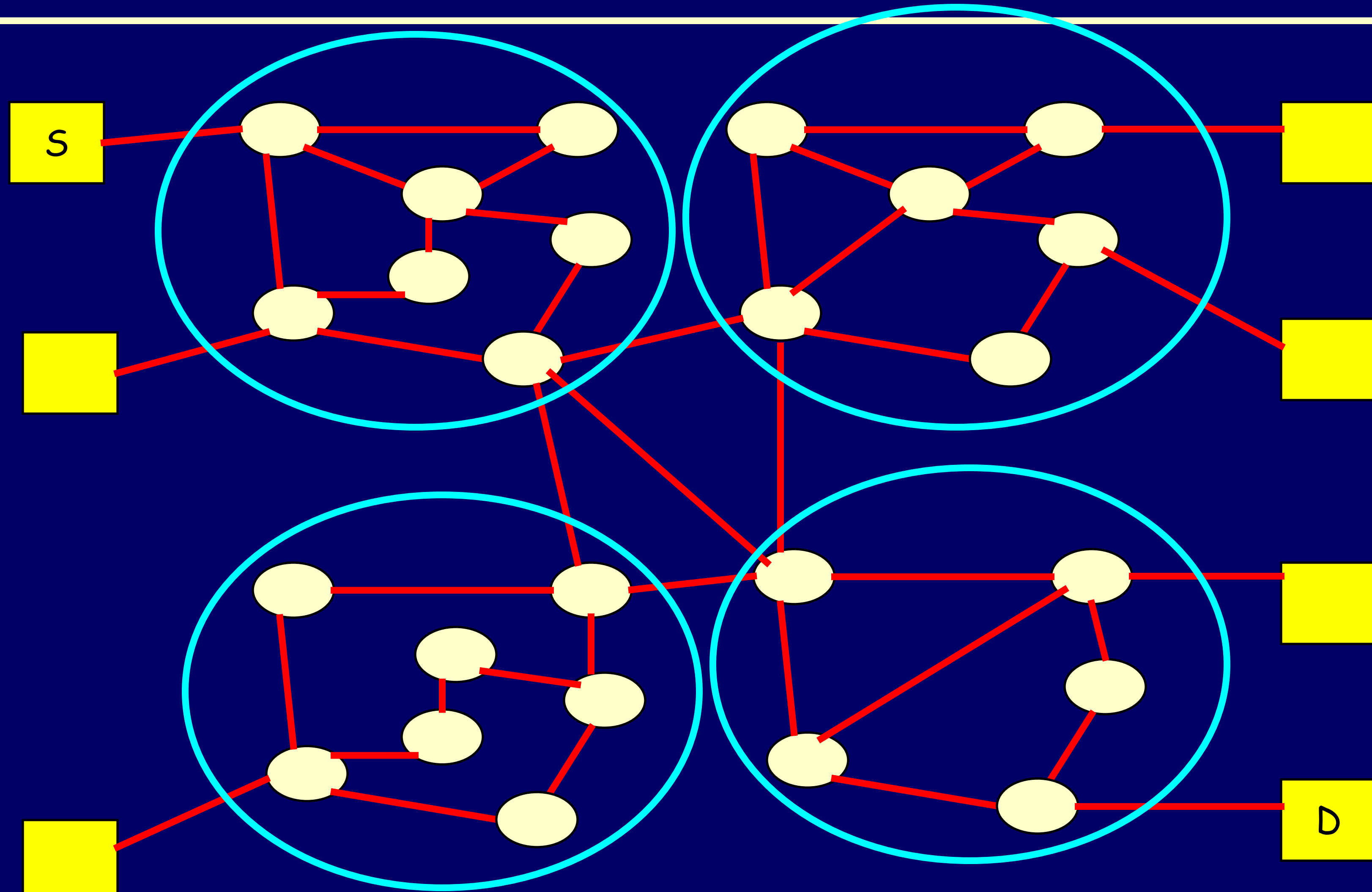


• The source has the entire network state information (cost, delay, etc.. of each edge)

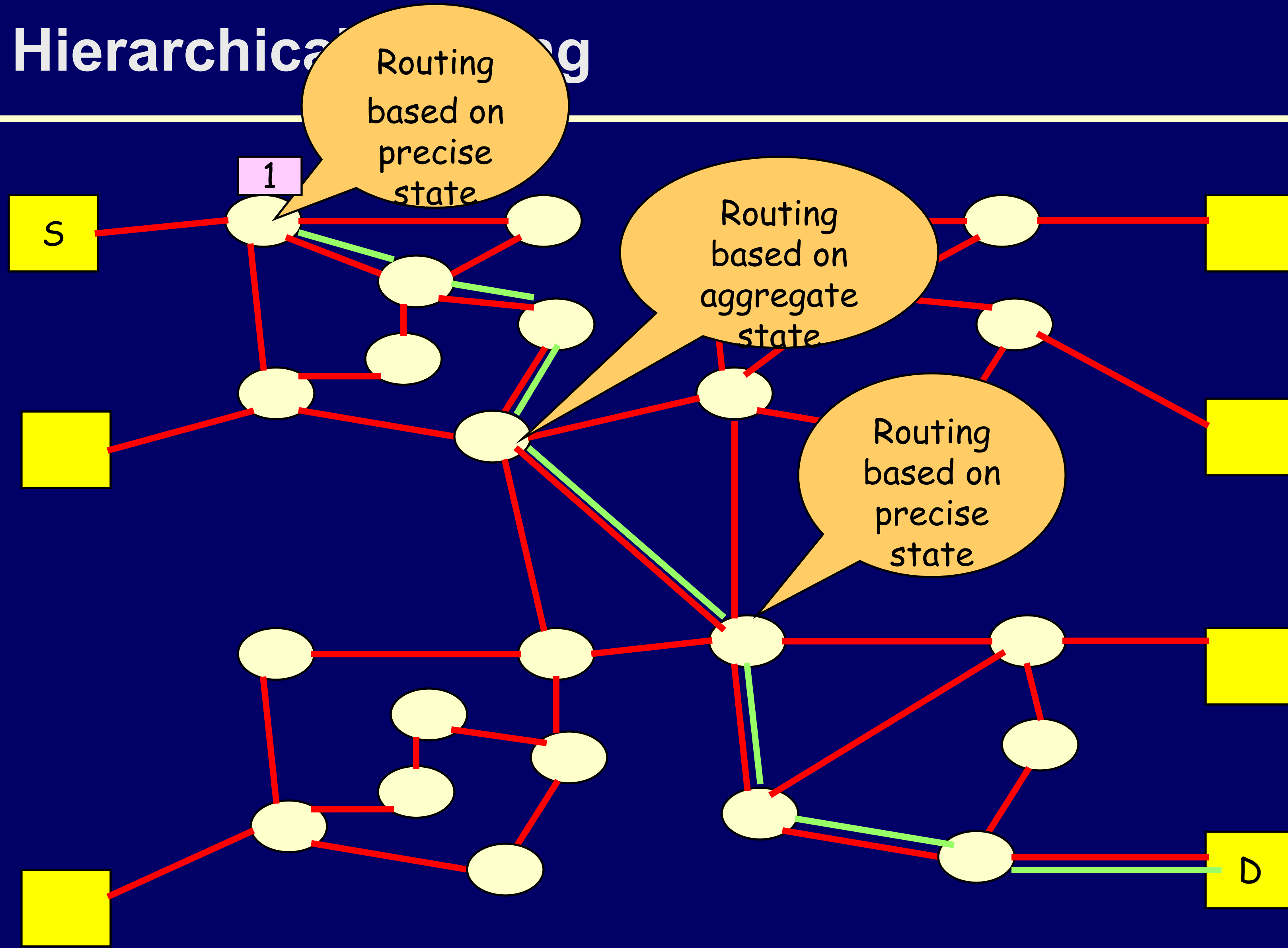
Distributed / Hop-by-hop Routing



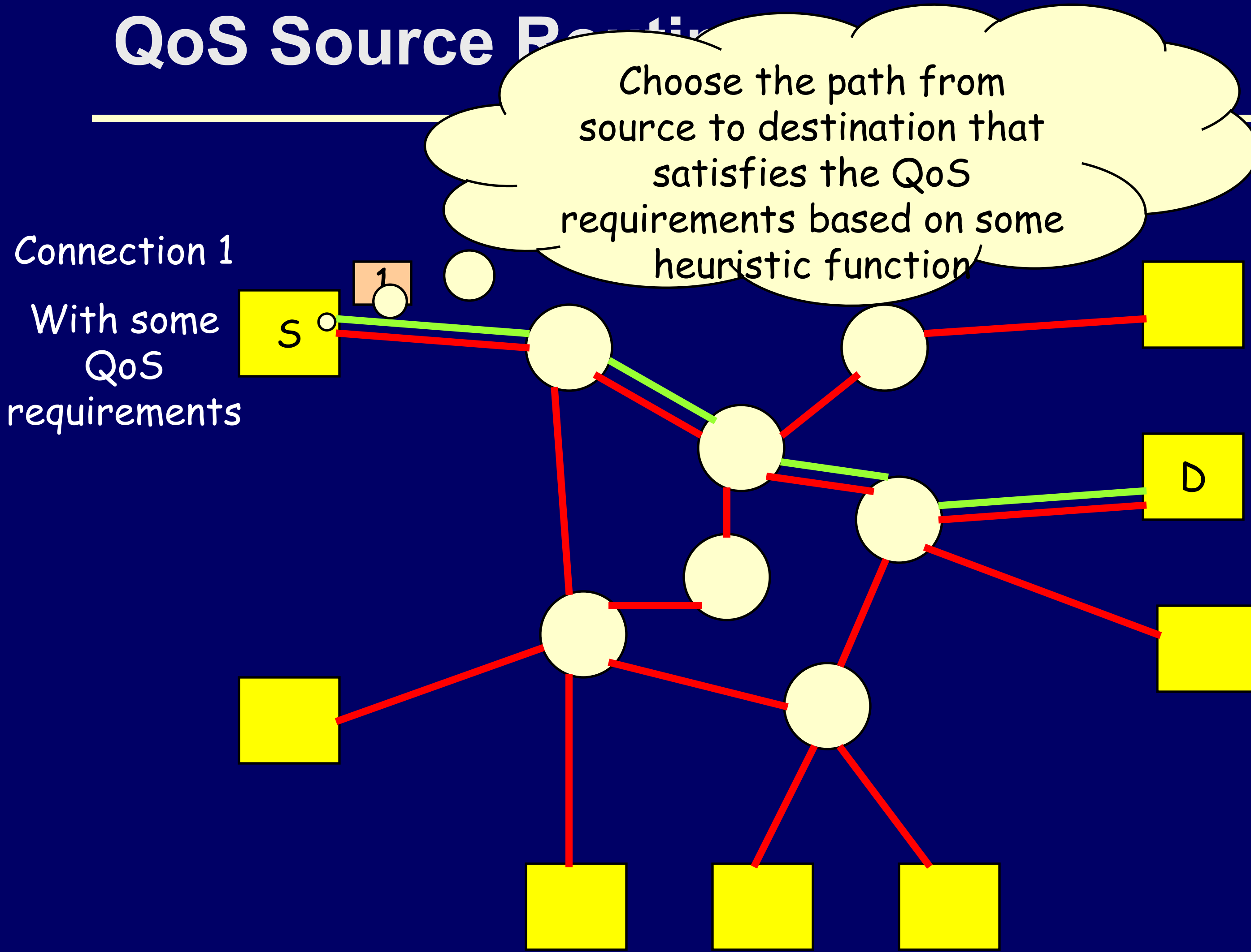
Hierarchical Routing



Hierarchica g

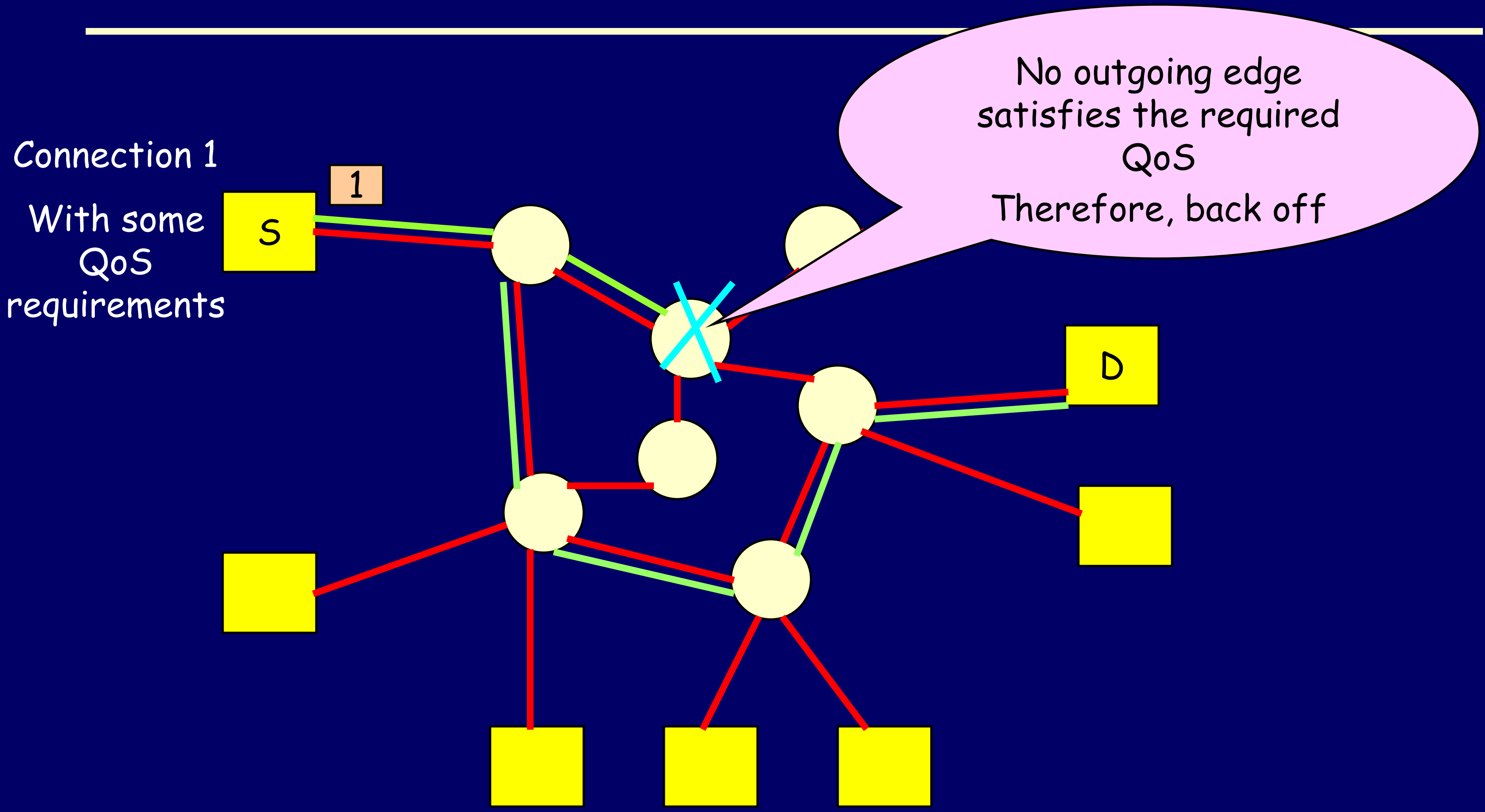


QoS Source R



- The source has the entire network state information (cost, delay, etc.. of each edge)

QoS Hop-by-hop Connection setup



Channel establishment – Resource reservation

- Once a feasible path is found, resources such as bandwidth, buffers are reserved along the path

- Resource Reservation Protocol (RSVP)
  - Receiver-driven protocol
  - Supports multicast as well as unicast
  - Provides different type of reservation styles